



NIMBUS - The Agro e-Duster



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Introduction

Nimbus is an autonomous electric crop duster that delivers unmatched spray precision and productivity, cutting operating costs lower than any fuel-powered alternative.

Design Requirements

Hopper Capacity: 200 liters
Endurance: 35 Minutes
Minimum Take-off Distance: 300m
Cruise velocity: 35 m/s
Working velocity: 30 m/s

Mission Profile



Figure: Mission Profile

Specifications

Horizontal Stabilizer

NACA 0012
Span: 3m
Cr= Ct :1m
Incidence: -1°

Wings

High Wing Configuration
Airfoil: NACA 2412
Span: 12m
MAC: 1.057m
Aspect Ratio: 11.429
Reference Area: 12.569 m^2
Ct: 0.9m Cr: 1.2m
Taper Ratio: 0.75
Incidence: 3°
Dihedral: 4°

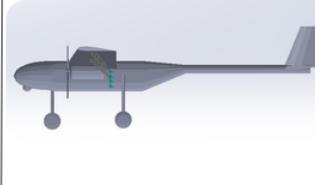
Vertical Stabilizer

NACA 2412
Span: 1.5m
Cr: 1.5m Ct: 0.9m
Sweep: 16.7°

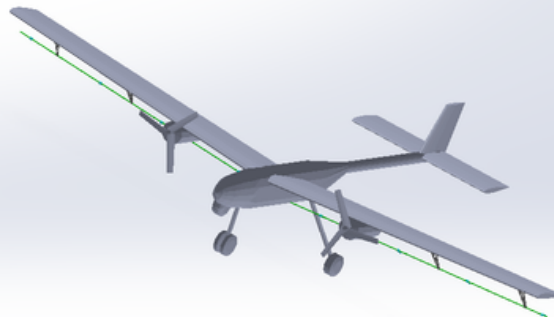
Aircraft Design



Top view



Side view



Isometric view

Aircraft Design



Conceptual Render

Configurations

General Specifications

Engine Power: $2 \times 45 \text{ hp}$
Swath Width: 14 m
Work Rate: Rs 5 per Ha
Total Coverage: 60 Ha

Weights

MTOW: 576 kg
Empty Weight: 293 kg
Battery Weight: 83 kg
Hopper Capacity: 200 ltrs

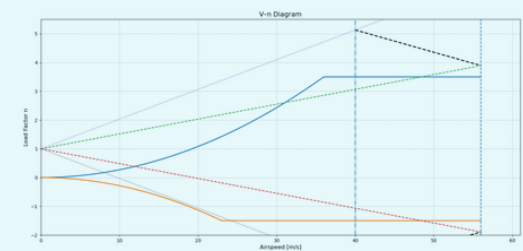
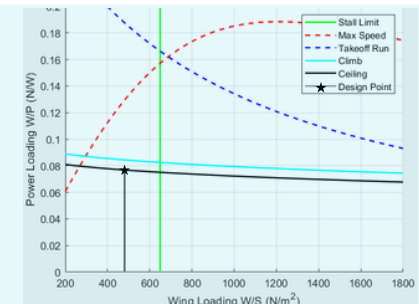
Performance

Take-off Distance: 460m
Landing Distance: 390m
Cruise Speed: 40m/s
Max. Speed: 48m/s
Cruise Altitude: 500m

Energy

Battery type: Li-on
Battery weight: 83kg
Energy Density: 265 wh/kg
Charging Rate: 210A
Discharging Rate: 325A

Matching plot and Flight Envelope



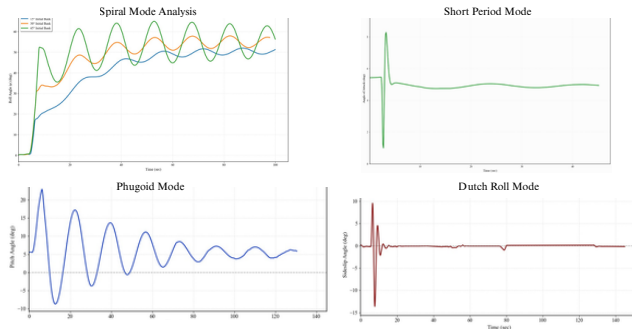


NIMBUS - The Agro e-Duster

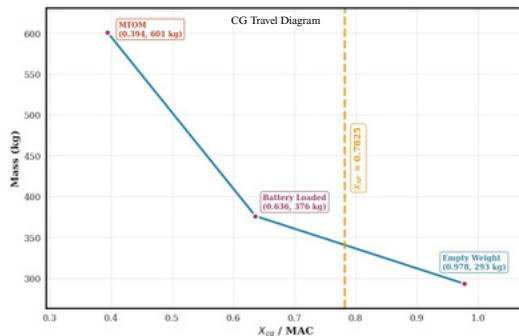


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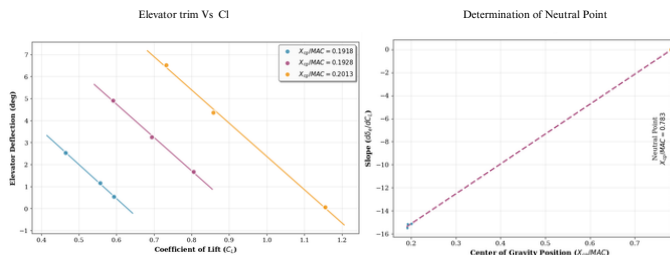
Dynamic Stability Test



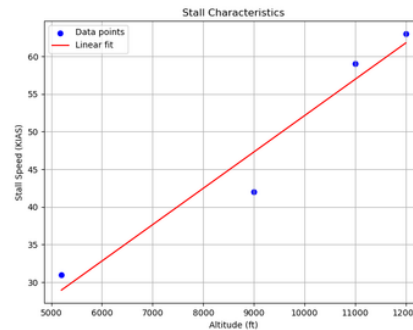
CG Travel Diagram



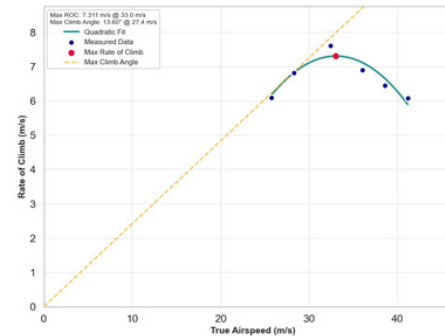
Neutral Point



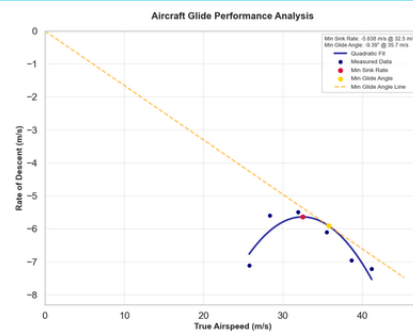
Stall Characteristics



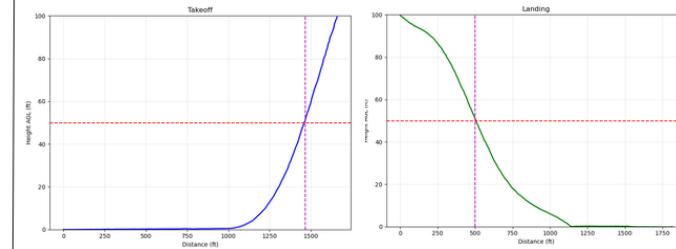
Climb Characteristics



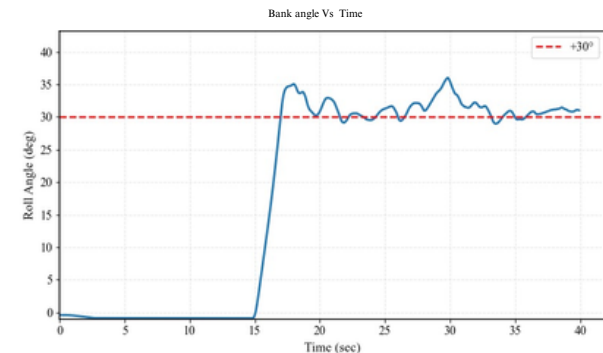
Descent Characteristics



Take-off and Landing Characteristics



Handling Quality Test



The Aircraft was rated 4 on the Modified Cooper–Harper (MCH) scale, indicating that the operator's effort is moderate, although the vehicle may exhibit drift in the presence of disturbances and requires pilot compensation.

Conclusion

The aircraft demonstrated reliable handling, stability, and adaptability in varied conditions, making it ideal for crop dusting. While there are many fuel powered giants for larger farm, but almost none for medium sized farm and Nimbus aims to fill the gap.